

ANDREI LEBEDEV

Machine Learning Engineer – RAG, LLM Alignment, LLM Systems

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EXPERIENCE

T-Bank, Enterprise AI Search, NLP/ML Engineer

Jul 2025 – Present (1 yr 1 mo)

- Built groundedness evaluation and source citation for RAG answers
- Designed and developed an implicit corporate knowledge graph for enterprise search optimization and agentic workflows
- Developed RAG evaluation pipelines covering search relevance, answer quality and citation quality; automated LLM-based relevance annotation for IR benchmarks
- Established the team's ML experiment workflow
- Integrated new functional blocks into search templates and improved ranking metrics

T-Bank, Invest AI Systems, NLP/ML Engineer

Jul 2024 – Jul 2025 (1 yr)

- Built a multi-agent LLM system for automatic unit-test generation to improve developer workflows
- Implemented an AI system (LLM + OCR + Qdrant) for extracting financial indicators from reports, saving analysts up to 20 hours per month
- Fine-tuned BERT on corporate-domain data and improved support-ticket clustering quality
- Delivered a workshop on AI agents to 200 participants

RESEARCH PROJECTS

Research Project: RL Methods for Adapting LLMs to Russian NER

Sep 2025 – May 2026

- Improved performance by +37% on NER tasks and +5% on average across benchmarks without significant quality degradation
- Built a GRPO training pipeline for Qwen3-4B on the NEREL dataset with a custom RLVR reward
- Built a comprehensive evaluation pipeline: target NER task, 30+ Russian-language benchmarks from LLMTF Open and RuArena comparisons
- Trained SFT baselines, showed GRPO advantages in this setup and analyzed overfitting
- Added filtering of uninformative generations inspired by DAPO

Bachelor Thesis: Post-Training Methods for LLMs in Russian

Sep 2024 – May 2025

- Improved key metrics without degrading general model capabilities: up to +18% on RU Arena Hard, +3% on LIBRA and +2% on Flores en→ru
- Studied LLM alignment and post-training methods: SFT, RLHF, DPO, SimPO and KTO
- Implemented a training and automatic evaluation pipeline across different data mixtures and methods
- Worked with SLURM on the MSU-270 compute cluster

Research Project: Automatic Code Generation with LLMs

Dec 2023 – May 2024

- Studied model behavior under quantization
- Reproduced research results on code-generation benchmarks
- Compared Russian and English code-generation quality using HumanEval and RU HumanEval

EDUCATION

Lomonosov Moscow State University, CMC Faculty

2025 – Present

- M.Sc. in Applied Mathematics and Computer Science

Lomonosov Moscow State University, CMC Faculty

2021 – 2025

- B.Sc. in Applied Mathematics and Computer Science
- GPA: 4.6/5

SKILLS

- **LLM/NLP:** RAG, agents, SFT, DPO, RL (GRPO, DAPO, RLHF), embeddings, reranking, LLM evaluation, Elasticsearch
- **Frameworks:** PyTorch, Transformers, TRL, PEFT, vLLM, Qdrant, Ray, TensorFlow, scikit-learn, Pandas, LangChain
- **Programming & Tools:** Python, C++, C, Docker, Git, Airflow, SLURM, Unix/Linux, PostgreSQL
- **Languages:** Russian (native), English (Upper-Intermediate)

COURSES

Practical RL, Yandex Data School; Modern NLP: Large Language Models, VK Education; LLM Pro, Deep School